
Pattern Arithmetic: Higher Truths from Assembly

Sandeep Lakshminarayan Chiluveru

September 25, 2026

Abstract

This is a second companion paper to *Pattern Arithmetic: Typed Algebra of Bounded Units on a Phase-Colored Sphere*, alongside *Pattern Arithmetic: Inference from Bowls and Paths*. The first defines assembly ($\bowtie$) and an enclosure in which units may descend an energy and rest as one camp or several; the second reads a bowl of evidence into a verdict, but never asks what a settled camp itself is entitled to be. This paper collects what is currently known about that question, and it is deliberately small. Two points are already closed, as direct consequences of choices the companion paper already made rather than of anything built here: the actor/patient question, settled at the level of a label rather than a difference in motion, and the absence of astrophysical mass segregation, a forced consequence of excluding volume from motion. One point is genuinely open — fine-grained multi-vector matching, where an atom’s own bundle, not its coarse derived vector, decides how it bonds. What this paper does not yet do is the harder thing its own title promises: say what a frozen camp, read and re-read at successive checkpoints, is entitled to be called, and whether that naming is itself a higher unit of truth or only a description of one. That is stated here as this paper’s own direction, not answered as its content.

1 Introduction

A camp settles. Something froze at a resolution η , with a body $\vec{P}_c$ and a membership — and the companion paper is explicit that the body is a description, measured after the fact, never a destination the motion sought. Read again at a finer η , or later in time, the same units may freeze into a different shape, or one pile where there were two: a camp is a checkpoint, not a fixed thing, and two checkpoints close enough to each other are read as the same freeze only by a further, named allowance (companion paper, Enclosure).

None of this says what a checkpoint *is*, in the sense a bowl of evidence becomes a verdict in the other companion paper. That is the question this paper exists to carry, separately from inference, so that it can grow on its own rather than as an appendix to a paper about something else.

What follows is not yet that answer. It is the ground already cleared: two questions the companion paper’s own axioms settle as a side effect, stated here plainly since neither was recorded at the time, and one question about how an atom’s inner structure governs assembly that remains genuinely open. The larger question — what a settled camp, read against known patterns or against its own history of checkpoints, is entitled to be named, and whether that naming closes into a symbol a higher unit could itself be built from — is named in the abstract and left there. Early exploration toward it (reading a settled camp against known references, and characterizing when such a reading is trustworthy rather than lucky) exists only as worked, unpublished calculation; none of it is claimed here as a result.

2 What Is Inherited

We use, without re-deriving, the following from the companion paper. Equation and axiom numbers refer to that paper.

- The atom $x = (r, \hat{n}, \chi, \tau) \in \mathcal{S}$, with $r \leq 1$ a normalized strength (Axiom 1).
- Assembly, $\bowtie$: two units related while both remain intact, $A \bowtie B = (A, B, \kappa_{AB}, U_{AB})$, with $U_{AB} < 0$ for nonzero strengths and never repelling (Axiom 6).
- The enclosure: external seats $\vec{P}$ descend the energy $\sum U_{ij}$ without mass or momentum until a stated force threshold η is crossed; a configuration below it is a *checkpoint*, not an exact equilibrium, since the pull between any two units of nonzero strength never reaches zero at finite distance (Enclosure, Rest and Camps).
- Informational vectors (Axiom 7): an atom may carry N capped arrows and one uncapped derived sum V_{derived} . Coarse assembly runs on the derived pair; the bundle’s own arrows are not consulted by U_{AB} unless named explicitly.
- *Levels are closed*: a camp carries no representation of what it constitutes, and the enclosure supplies no comparison by which a checkpoint could name itself.

3 What Assembly Already Settles

Two settled points are worth recording plainly, since both follow from Axiom V4’s exclusion of $\mathcal{V}_{\text{atom}}$ from motion (companion paper, Enclosure § Rest) rather than from anything built here.

The actor/patient question. Closed at the level of a label: polarity ($|r_A - r_B|$) distinguishes a mutual bond from a one-sided one, and no further behavioral mechanism is missing — one was tested directly and found to require exactly the volume-weighted motion that exclusion rules out on stability grounds, so the honest final scope of this question is a description of the bond, not a difference in how it moves.

Mass segregation. The astrophysical phenomenon — stronger members sinking toward a group’s centre — was named early in the companion paper’s own development as an expected consequence, while its dynamics was still built around a mass-like quantity. Once motion settled as excluding $\mathcal{V}_{\text{atom}}$ entirely, that mechanism no longer applies here, confirmed directly: equal force with nothing dividing it gives identical movement regardless of strength or volume. The astrophysical phenomenon keeps its real name; nothing in this framework currently reproduces it, since the quantity that would drive it was deliberately kept out of motion.

4 Open: Fine-Grained Multi-Vector Matching

Coarse assembly runs on the derived vector (companion paper, Axiom 7), so U_{AB} never needed to generalize; simultaneous multi-partner bonding via informational vectors directly remains optional, with an unconstrained double sum and an exclusive (Hungarian) matching identified as the two candidates. They agree whenever alignment quality is roughly uniform across candidates, but rank-order pairing can be strictly worse than optimal when strength and alignment pull in different directions — verified with a constructed adversarial case (-1.17 for rank order versus -1.68 true optimum).

5 Scope and the Question This Paper Is For

This paper is small on purpose. It records what the companion paper’s own choices already settle about assembly, once, so that neither has to be rediscovered; and it names one open question about how an atom’s own bundle should govern multi-partner bonding, where the coarse derived-vector treatment is known to be incomplete.

What it does not do is name a checkpoint. A frozen camp has a body and a membership and, by the companion paper’s own accounting, nothing else — no shape, no representation of what produced it, no claim on what it should be called. Whether a checkpoint can be read against known patterns and trusted, whether a sequence of checkpoints under an allowance converges on something worth naming, and whether such a naming is a genuine higher unit of truth or only a compact description of a lower one — these are the paper’s own stated direction, not its content. They are left here, deliberately, for the work this paper exists to carry.

Aside: A Note Parked Here for the Companion Paper

This belongs beside *Pattern Arithmetic*’s own Thesis, not here — that paper is frozen for now, so it is held in this one until it migrates.

Not every axiom is earned the same way. Some are forced: π ($3.14159265\dots$) and e ($2.71828182\dots$) carry no freedom at all, falling out of their own definitions identically regardless of who does the deriving — discovered, not invented, in the strongest sense available. Some are genuinely invented, the way physics’ free parameters are: the fine-structure constant, the particle masses, Planck’s constant’s numerical value — structure discovered, specific value measured, with no known reason it could not have come out otherwise. And some, like the companion paper’s own $r \leq 1$, are neither: a normalization,

a choice of scale, cheap to invent because no structural weight rides on the specific number — rescale it and nothing about the theory’s shape changes.

The test that actually sorts these, worth stating plainly: not whether a rule was discovered or invented, but whether a different choice would have produced a different theory. π fails this test by having no alternative to compare against. $r \leq 1$ fails it by mattering only as a scale. Mix as phasor summation, or Gather’s idempotence, pass it — a different choice there is a different algebra, not the same one in different clothes. This is the reasoning underneath the companion paper’s own line, that it “states its own rules as axioms rather than discovering them”; it is the same standard this paper’s own closure axioms, wherever they are eventually stated, should be held to.